

Math block sampler

This file exercises both block-level math ($...$ and fenced “math blocks) and inline math ($...$). Real LaTeX math in every block.

An inline atom: the golden ratio $\varphi = (1 + \sqrt{5}) / 2$ sits in running text. The fine-structure constant is approximately $\alpha \approx 1/137$.

A definite integral:

$$\int_{-\infty}^{\infty} e^{-x^2} \, dx = \sqrt{\pi}$$

A sum, in a fenced math block:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

A simple fraction and a binomial coefficient on one line:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \quad \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

A two-by-two matrix and its determinant:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

An aligned derivation:

$$\begin{aligned} (a+b)^2 &= a^2 + 2ab + b^2 \\ (a-b)^2 &= a^2 - 2ab + b^2 \\ (a+b)(a-b) &= a^2 - b^2 \end{aligned}$$

The Pythagorean identity: $a^2 + b^2 = c^2$.